PHY202 (Overview of Mathematical Methods): 1st Mid-Sem Exam

Duration: **60** minutes; Maximum marks: **20**

1. Find the real and imaginary parts of the following functions:

   (i) $f(z) = (1+2i)z^2 + iz$, (ii) $f(z) = e^{iz}$.

   [2+2]

2. Check whether or not the following functions satisfy the Cauchy-Riemann conditions:

   (i) $f(z) = 1/z$, (ii) $f(z) = \dfrac{2z-i}{iz+2}$.

   [3+3]

3. Find the first two non-zero terms in the Laurent series of the function $g(z) = \dfrac{\cos z}{(2z-\pi)^2}$ about $z_0 = \pi/2$. What is the values of the integral $\oint_C g(z)dz$ if C is the unit circle $|z| = 1$?

   [3+1]

4. Use your knowledge of functions of complex variables to evaluate I_1 and I_2 defined below:

   $$I_1 = \int_0^{2\pi} \frac{\cos\theta}{(5-4\cos\theta)}\, d\theta$$

   $I_2 = \oint_C \dfrac{e^{3z}}{(z-\ln 2)^4} dz$, where C is the square with vertices at ± 1, $\pm i$

   [3+3]